

Single cell sequencing – practical guidelines

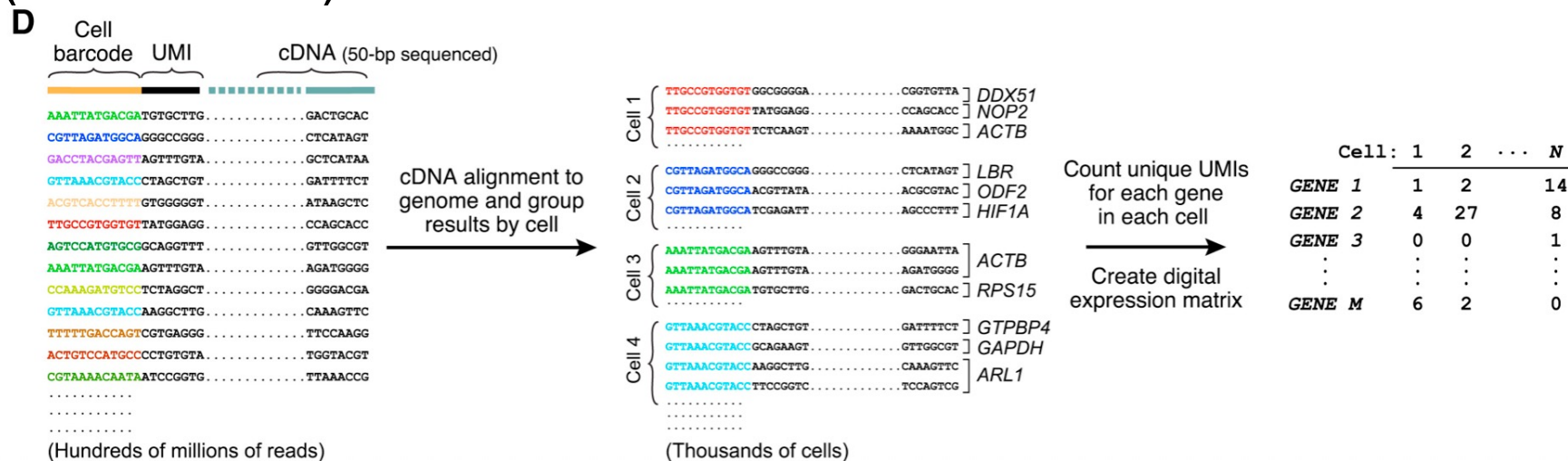
Neuro-genomics
Exercise 8

Single cell analysis using Seurat (using R)

- Standard pre-processing workflow
- Normalizing the data
- Identification of highly variable features (feature selection)
- Scaling the data
- Perform linear dimensional reduction
- Determine the 'dimensionality' of the dataset
- Cluster the cells
- Run non-linear dimensional reduction (UMAP/tSNE)
- Finding differentially expressed features (cluster biomarkers)
- Assigning cell type identity to clusters

Single cell analysis using Seurat

- The raw data:
Count matrix: the values in this matrix represent the number of molecules for each feature (i.e. gene; row) that are detected in each cell (column).
- Note that in addition to 3'-end sequencing with unique molecular identified (UMI), there are protocols for full-length mRNA sequencing (without UMI)



Single cell analysis using Seurat

- Standard pre-processing workflow

The number of unique genes detected in each cell (or the total number of molecules detected within a cell):

- Low-quality cells or empty droplets will often have very few genes
- Cell doublets or multiplets may exhibit an aberrantly high gene count

The percentage of reads that map to the mitochondrial genome (not needed in all applications):

Low-quality / dying cells often exhibit extensive mitochondrial contamination

Single cell analysis using Seurat

- Normalizing the data

Normalization between the individual cells (note that this step is probably not needed for in situ-based single cell sequencing):

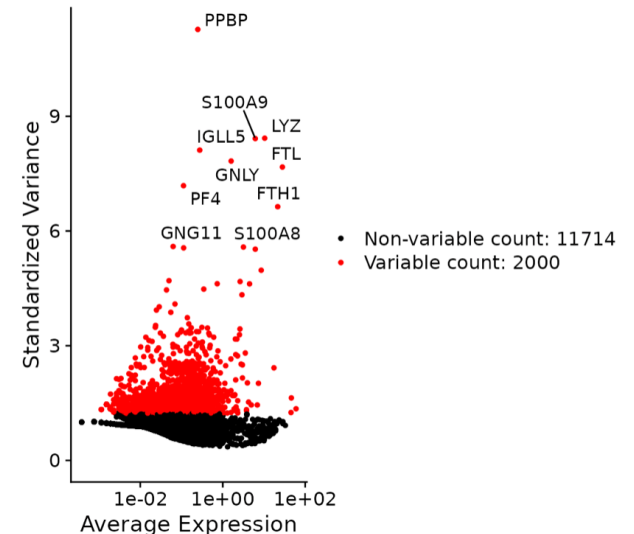
- normalize the expression measurements for each cell by the total expression
- multiply this by a scale factor (for example 10,000)
- log-transform the result

Single cell analysis using Seurat

- Identification of highly variable features (feature selection)

The motivation: instead of using all the dimensions (genes), focus on the genes that are most likely to separate the cells:

- Highly variable genes = genes that have high variance compared to gene with similar average expression level (highly variable genes are highly expressed in some cells, and lowly expressed in others)
- Choose genes according to previous knowledge



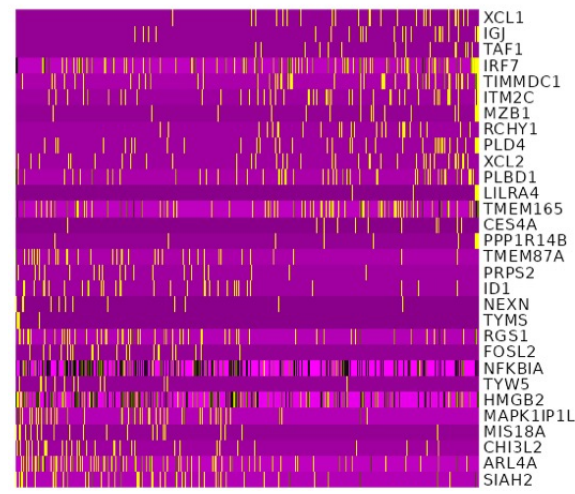
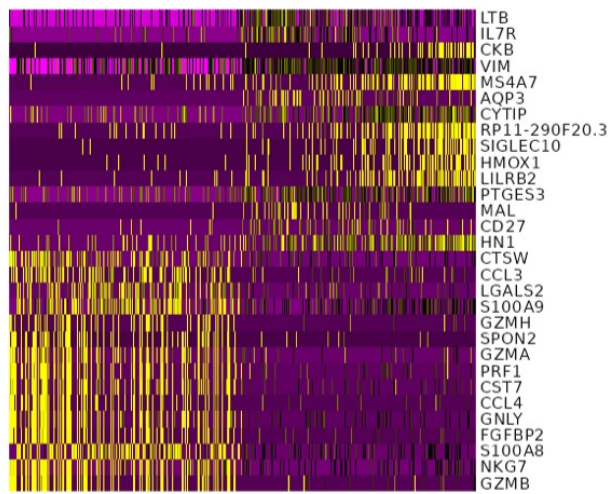
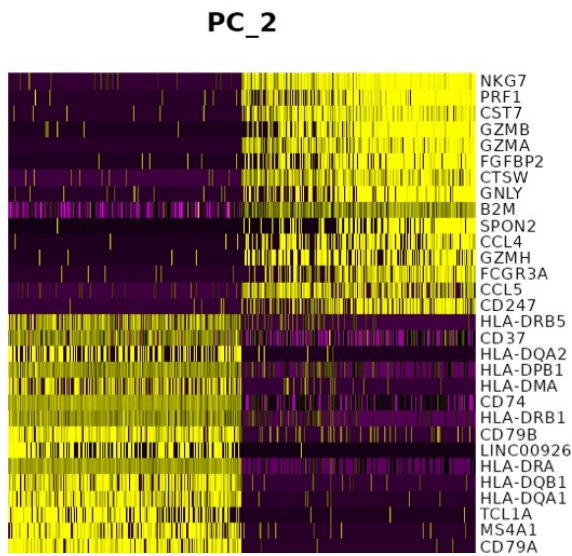
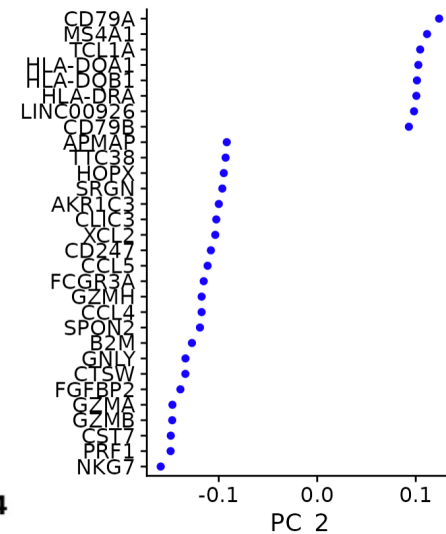
Single cell analysis using Seurat

- Scaling the data (note that this step is aggressive, and therefore might not be a good fit for all experiments, especially for experiments with low number of features (genes)):
 - Shifts the expression of each gene, so that the mean expression across cells is 0
 - Scales the expression of each gene, so that the variance across cells is 1
 - This step gives equal weight in downstream analyses, so that highly-expressed genes do not dominate

Single cell analysis using Seurat

- Perform linear dimensional reduction:
 - Perform principal components analysis (PCA)
 - Explore which genes are included in each PC
 - Explore the cells with the highest/lowest expression of the genes in each PC

Yellow = highest; Purple = lowest; Black = 0

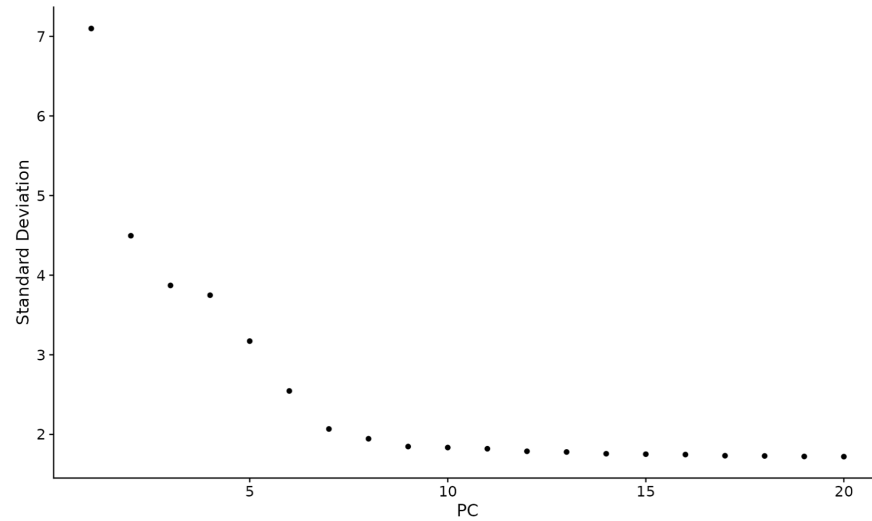


Single cell analysis using Seurat

- Determine the ‘dimensionality’ of the dataset:

The top principal components represent a robust compression of the dataset. However, how many components should we choose to include? 10? 20? 100?

- The explorations describe before can be useful
- ‘Elbow plot’ = the percentage of variance explained by each PC



Single cell analysis using Seurat

- Determine the ‘dimensionality’ of the dataset:
 - Permutation analysis

Generate X randomizations of the data such that within each realization, the values along every row (gene) of the scaled expression matrix are randomly permuted. This operation is leaving the expression distribution unchanged. PCA is then performed on each of these X “randomized” datasets. Significant PCs in the **un-permuted** data are identified as those with larger eigenvalues compared to the highest eigenvalues across the X **randomized** datasets

Permutation analysis:

Compare that score that each PC gets using real data, to that obtained using permuted (non-biological) data. This allows to compute a p-value for each PC

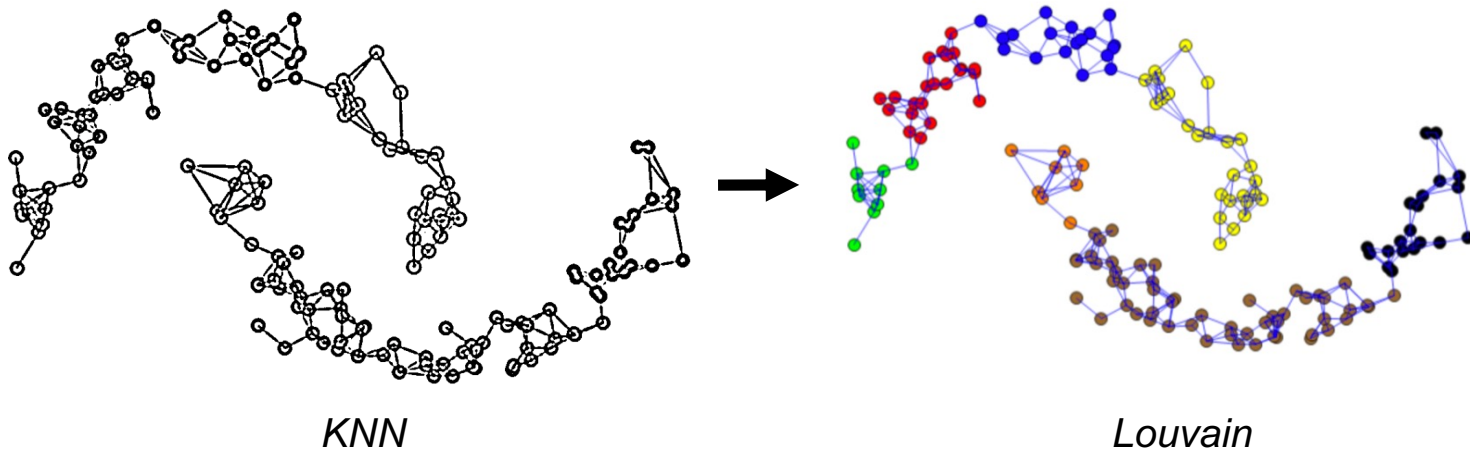
PC: p-value

- PC 1: 9.12e-161
- PC 2: 1.47e-114
- PC 3: 1.98e-33
- PC 4: 9.49e-58
- PC 5: 5.05e-56
- PC 6: 2.76e-55
- PC 7: 9.76e-23
- PC 8: 1.48e-11
- PC 9: 5.22e-12
- PC 10: 6.33e-08
- PC 11: 3.75e-08
- PC 12: 0.000172
- PC 13: 0.00438
- PC 14: 0.133
- PC 15: 0.479

Single cell analysis using Seurat

- Cluster the cells:
 - Calculate Euclidean distances in the PCA space
 - Embed cells in a graph structure. This is done using a K-nearest neighbor (KNN) graph, with edges drawn between cells with similar feature expression patterns (images from [here](#))
 - Partition this graph into highly interconnected 'communities' using the [Louvain algorithm](#)

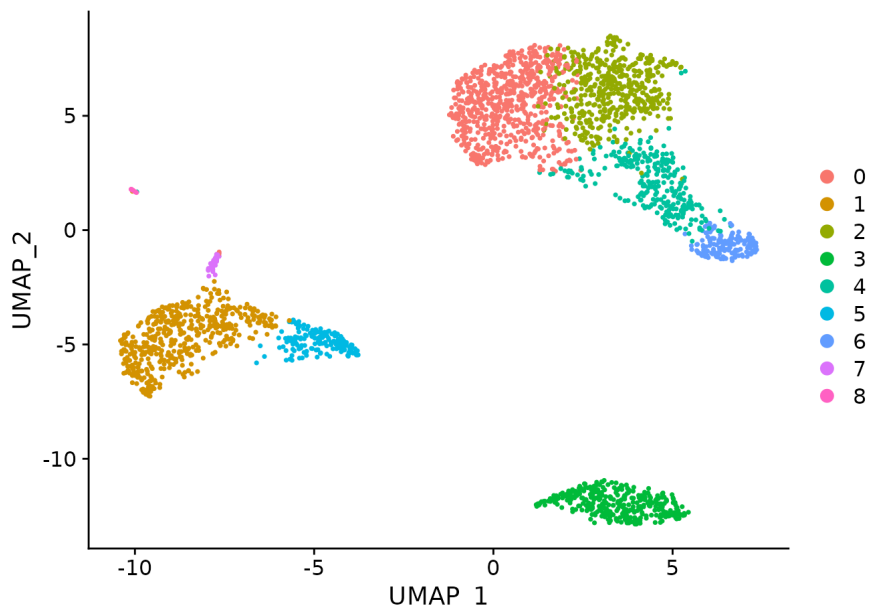
KNN = Given a positive k , cell i is connected to cell j if cell j is among the k -nearest neighbors of cell i (up to a certain threshold of distance)



Single cell analysis using Seurat

- Run non-linear dimensional reduction:
 - tSNE or UMAP can be used to visualize and explore cells
 - The goal of these algorithms is to learn the data in order to place similar cells together in a low-dimensional space
 - As input to UMAP or tSNE, use the same PCs that were input to the clustering analysis

Cells within the graph-based clusters determined above (KNN+Louvain) should co-localize on these dimension reduction plots



Because UMAP and tSNE are non-linear, and hard to control (for example, they are very sensitive to parameters), do not use them for clustering! Use them only to visualize the data in 2D space

Single cell analysis using Seurat

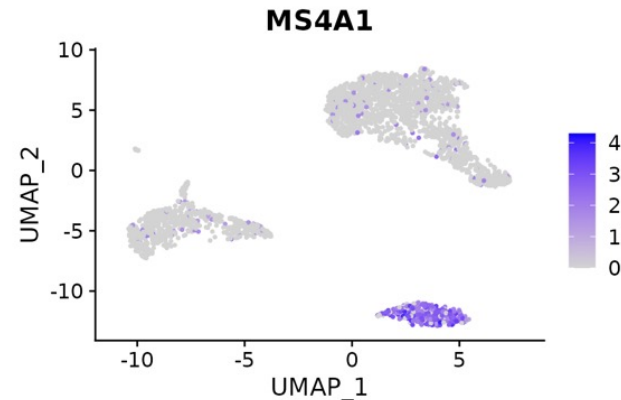
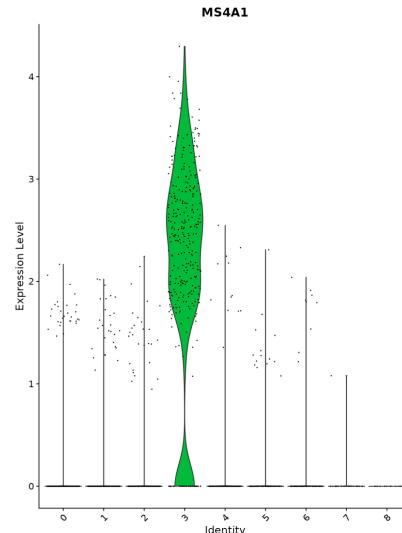
- Finding differentially expressed features (cluster biomarkers):
 - Internal filtering can be performed, which means that not all of the quantified genes will actually be tested for differential expression (DE)
 - For example, apply threshold for percentile (in either samples) and/or threshold for fold change

##		p_val	avg_log2FC	pct.1	pct.2	p_val_adj
##	FCGR3A	8.246578e-205	4.261495	0.975	0.040	1.130936e-200
##	IFITM3	1.677613e-195	3.879339	0.975	0.049	2.300678e-191
##	CFD	2.401156e-193	3.405492	0.938	0.038	3.292945e-189
##	CD68	2.900384e-191	3.020484	0.926	0.035	3.977587e-187
##	RP11-290F20.3	2.513244e-186	2.720057	0.840	0.017	3.446663e-182

pct = percentile (0 to 1)

adj = adjusted p-value, based on Bonferroni correction using all genes

Violin plot



Single cell analysis using Seurat

- Finding differentially expressed features (cluster biomarkers):
- Which tests can be done?
 - “bulk RNA-seq analysis methods do not generally perform worse than those developed specifically for scRNA-seq” from [here](#)
 - Likelihood ratio test assuming an underlying Poisson distribution
 - Likelihood ratio test assuming an underlying negative binomial distribution
 - “DESeq2”: DE based on a model using the negative binomial distribution
 - How can we tell which model to choose?

Single cell analysis using Seurat

- Finding differentially expressed features (cluster biomarkers):
- Which tests can be done?

Because we are comparing tens (or more) of single cells to tens (or more) of single cells, maybe we can use:

- t-test, but it is less recommended because the assumptions are:
 - The two samples are independent of one another
 - The two populations have equal variance or spread
 - The two populations are normally distributed
- Wilcoxon rank sum test
 - Similar to t-test, but without assumption regarding normal distribution (the test is non-parametric)

Single cell analysis NOT using Seurat

- Finding differentially expressed features:
- Do we need to cluster before the differentially expression calculation?

Usually we either:

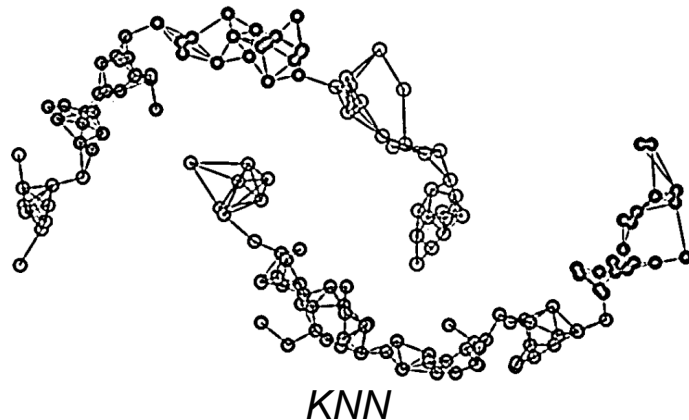
a) compare single cells from one cell type to single cells from another cell type (inside the same sample),

or

b) compare single cells from one cell type in condition A (say the control sample) to single cells from the same cell type in condition B (say the test sample).

However, these comparisons are based on the clustering of cells into cell type, which is a non trivial process that is not without errors.

- The alternative is to perform the analysis on the KNN graph as in [here](#) and [here](#)
- The idea is to look for 'neighborhoods' in the KNN graph which are different between the two conditions
- This analysis can also be performed within a sample



Single cell analysis using Seurat

- Assigning cell type identity to clusters:
 - Use canonical markers to match the clustering biomarkers to known cell types
 - Prior knowledge is needed about cell types markers

Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
1	CD14, LYZ	CD14+ Mono
2	IL7R, S100A4	Memory CD4+
3	MS4A1	B
4	CD8A	CD8+ T
5	FCGR3A, MS4A7	FCGR3A+ Mono
6	GNLY, NKG7	NK
7	FCER1A, CST3	DC
8	PPBP	Platelet

